

Spatial Analysis of LMPs and Generation Siting

2

UNESP — LaPSEE Graduate Minicourse

Session 2 — 2 h

- I. What Session 1 gave us — and what it didn't
- II. **Prices from the network**: duals, made concrete
- III. **The spatial weight matrix $\mathbf{W}$** (new today)
- IV. **Detecting dependence**: Moran & LISA (new today)
- V. **Explaining dependence**: spatial regression
- VI. DG siting as inverse regression
- VII. Reality check on RTS-GMLC + wrap-up

8'

25'

15'

15'

25'

15'

12'

Take-away

The LMP is the *dual variable* of the DC-OPF nodal balance (Schweppe et al. 1988). Its spatial structure is *mechanical*, not statistical. Parts III–V build the spatial toolkit we did not reach last time; Parts II and VI are where the power system enters.

I. What Session 1 gave us — and what it didn't

These objects were built and checked last time — we *use* them today:

- **DC power flow.** $P^{\text{inj}} = \mathbf{B}\theta$ with the susceptance Laplacian $\mathbf{B} = \mathbf{L}_Y$; slack reduction; branch flows $f_{ij} = b_{ij}(\theta_i - \theta_j)$.
- **PTDF.** $\mathbf{H} = \mathbf{B}_d \mathbf{A}_r \mathbf{B}_{-s,-s}^{-1}$. Row ℓ tells you how an injection anywhere loads line ℓ . We saw a PTDF column *is* a current divider.
- **Spectrum of the Laplacian.** $\lambda_2 =$ algebraic connectivity; the Fiedler vector; effective resistance $Z_{\text{eff}}(i, j)$ as electrical distance.
- **Centrality.** Topological vs. electrical betweenness — and why topology alone misleads on real grids.

Take-away

We have the *structural* / *circuit* half of the picture.

Session 1 ran out of time before its spatial half. So today is *self-contained* for the statistics:

- **Spatial weights $\mathbf{W}$** — a declared notion of “neighbour” on the grid (Part III).
- **Moran's I and LISA** — how to *detect* and *localise* spatial dependence in bus-indexed data (Part IV).
- **Spatial regression (SAR / SEM / SDM)** — how to *explain* that dependence, control for covariates, and read off spillovers (Part V).

Key idea. We will not present this as abstract econometrics. Every spatial object is anchored on the same circuit objects above: $\mathbf{W}$ comes from Z_{eff} and $\mathbf{H}$; the regression multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ is the statistical twin of the network reach $\mathbf{B}^{-1}$ we already met.

Why do two adjacent buses on the same grid see *different* energy prices?

- In an uncongested, lossless grid: they don't. *One* uniform clearing price.
- With **binding line constraints**, the cost of serving the next MW of demand depends on *where* that demand sits in the network.
- That location-dependent price is the **locational marginal price** LMP_i — the subject of this session.

Key idea. The LMP is not a market gadget bolted onto physics. It is the **shadow price** of the network's own optimisation. We will *remind* ourselves what a shadow price is, compute the LMPs of a 3-bus grid *by hand*, and only then look at a benchmark.

II. LMPs from KKT — primal, dual

Why do two adjacent buses on the same grid see *different* energy prices?

- Uncongested, lossless grid: they don't. *One* uniform price.
- With **binding line constraints**, the cost of serving the next MW of demand depends on *where* that demand sits.
- That location-dependent price is the **locational marginal price** LMP_i .

Key idea. The LMP is the **dual variable** of the DC-OPF nodal balance (Schweppe et al. 1988; Hogan 1992). Plan for Part II: write the *primal* LP as you know it, *build the dual / KKT*, read off the LMP, and then do the whole thing *by hand* on a 3-bus toy.

Step 1 — the DC-OPF primal (as you know it)

Decision variable: generation $P^g \in \mathbb{R}^{n_g}$; demand P^d fixed. Convex cost $C_g(P_g) = a_g + b_g P_g + \frac{1}{2} c_g P_g^2$.

Reduced (slack-eliminated, PTDF) form:

$$\begin{aligned} \min_{P_g} \quad & \sum_g C_g(P_g) \\ \text{s.t.} \quad & \mathbf{1}^\top P^g - \mathbf{1}^\top P^d = 0 \quad (\lambda) \text{ system balance} \\ & + \mathbf{H} \Delta \mathbf{P} \leq \mathbf{F}^{\max} \quad (\mu^+ \geq \mathbf{0}) \text{ upper line limits} \\ & - \mathbf{H} \Delta \mathbf{P} \leq \mathbf{F}^{\max} \quad (\mu^- \geq \mathbf{0}) \text{ lower line limits} \\ & P_g^{\min} \leq P_g \leq P_g^{\max} \quad (\underline{\nu}, \bar{\nu} \geq \mathbf{0}) \end{aligned}$$

with $\Delta \mathbf{P} = \mathbf{P}^{g,\text{net}} - \mathbf{P}^{d,\text{net}}$ the nodal net injection. (Equivalent to the $\mathbf{B}\theta = P^{\text{inj}}$ form of Session 1, with θ eliminated.)

Take-away

The **PTDF** you built last week is the **line-limit constraint matrix**. Same matrix, two roles: power flow $\rightarrow$ market.

Step 2 — build the dual: Lagrangian and KKT

Collapse $\mu_\ell \equiv \mu_\ell^+ - \mu_\ell^-$ and write the Lagrangian (gen-limit terms suppressed for interior units):

$$\mathcal{L} = \sum_g C_g(P_g) - \lambda(\mathbf{1}^\top P^g - \mathbf{1}^\top P^d) + \sum_\ell \mu_\ell (\sum_k H_{\ell,k} \Delta P_k - F_\ell^{\max}).$$

Stationarity at an *interior* generator g (so $\nu = 0$):

$$\frac{\partial \mathcal{L}}{\partial P_g} = C'_g(P_g) - \lambda + \sum_\ell \mu_\ell H_{\ell,g} = 0 \implies C'_g(P_g) = \lambda - \sum_\ell \mu_\ell H_{\ell,g}.$$

Complementary slackness: $\mu_\ell \geq 0$, and $\mu_\ell > 0$ *only* when line ℓ binds. Quadratic cost + linear constraints $\implies$ KKT are necessary *and* sufficient.

Take-away

The marginal cost of an interior unit equals $\lambda - \sum_\ell \mu_\ell H_{\ell,g}$. That right-hand side *is* the price seen at its bus.

The LMP at bus i is the marginal cost of serving one more MW of load there, $\text{LMP}_i = -\partial \mathcal{L}^* / \partial P_i^d$:

$$\text{LMP}_i = \underbrace{\lambda}_{\text{energy (uniform)}} - \underbrace{\sum_{\ell} \mu_{\ell} H_{\ell,i}}_{\text{congestion (locational)}} .$$

- **Energy** λ : the system balance dual — identical at every bus.
- **Congestion**: the binding-line duals μ_{ℓ} weighted by *your* PTDF on those lines.
- Uncongested $\Rightarrow$ all $\mu_{\ell} = 0 \Rightarrow$ one price everywhere.

Take-away

The spatial fingerprint of price is a weighted sum of PTDF columns of the binding lines — nothing else.

Step 1'' — the network-flow PRIMAL (standard pattern)

Let $\mathbf{A} \in \{0, \pm 1\}^{n \times m}$ be the **node-arc incidence** ($A_{i\ell} = +1$ if line ℓ leaves bus i , -1 if it enters), $\mathbf{f} \in \mathbb{R}^m$ the line flows, $\mathbf{B}_d = \text{diag}(b_\ell)$.

$$\begin{aligned} \min_{P^g, \mathbf{f}, \boldsymbol{\theta}} \quad & \sum_g C_g(P_g) \\ \text{s.t.} \quad & P_i^g - P_i^d = (\mathbf{A}\mathbf{f})_i, \quad \forall i \quad (\lambda_i) \text{ \textbf{nodal balance}} \\ & \mathbf{f} = \mathbf{B}_d \mathbf{A}^\top \boldsymbol{\theta} \quad (\boldsymbol{\eta}) \text{ DC coupling (KVL)} \\ & -\mathbf{F}^{\max} \leq \mathbf{f} \leq \mathbf{F}^{\max} \quad (\boldsymbol{\mu}^-, \boldsymbol{\mu}^+ \geq \mathbf{0}) \text{ capacity} \\ & P_g^{\min} \leq P_g \leq P_g^{\max} \quad (\underline{\nu}, \bar{\nu} \geq 0) \end{aligned}$$

This is exactly the canonical network-flow LP: **cost + incidence-balance + box bounds**. The only power-systems extra is the DC-coupling row tying $\mathbf{f}$ to angles (Kirchhoff's voltage law).

Take-away

One balance row per bus $\Rightarrow$ one dual λ_i per bus. That vector of node potentials is the LMP — no post-processing.

Step 2'' — the network-flow DUAL: node potentials

Lagrangian (gen-limit terms suppressed, interior units):

$$\mathcal{L} = \sum_g C_g(P_g) - \sum_i \lambda_i (P_i^g - P_i^d - (\mathbf{A}\mathbf{f})_i) - \boldsymbol{\eta}^\top (\mathbf{f} - \mathbf{B}_d \mathbf{A}^\top \boldsymbol{\theta}) + \boldsymbol{\mu}^{+\top} (\mathbf{f} - \mathbf{F}^{\max}) + \boldsymbol{\mu}^{-\top} (-\mathbf{f} - \mathbf{F}^{\max}).$$

Stationarity, the three KKT blocks:

$$\underbrace{C'_g(P_g) = \lambda_{n(g)}}_{\partial/\partial P_g: \text{price} = \text{local marginal cost}}$$

$$\underbrace{\eta_\ell = (\lambda_{\text{fr}(\ell)} - \lambda_{\text{to}(\ell)}) + (\mu_\ell^+ - \mu_\ell^-)}_{\partial/\partial f_\ell: \text{price drop across a line}}$$

$$\underbrace{\mathbf{A} \mathbf{B}_d \boldsymbol{\eta} = \mathbf{0}}_{\partial/\partial \boldsymbol{\theta}: \text{dual KVL}}$$

- $\text{LMP}_i = \lambda_i$: the price is the node potential itself.
- Interior generator $\Rightarrow$ its bus price equals its own marginal cost — read straight off the cost curve.
- On an *uncongested* line ($\mu_\ell^\pm = 0$) the dual KVL ties the λ_i together; a *binding* line injects μ_ℓ and lets adjacent prices **jump**.

Take-away

Same KKT, different bookkeeping: here congestion lives in $\boldsymbol{\eta}$ and the dual KVL, and the LMP needs no PTDF assembly.

Both roads, one price — node potentials $\equiv$ PTDF form

The two duals are the *same* object viewed in two bases. Eliminate θ and f from the node–arc KKT and you recover the boxed formula of Step 3:

$$\lambda_i = \underbrace{\lambda}_{\text{slack-bus / energy}} - \underbrace{\sum_{\ell} \mu_{\ell} H_{\ell,i}}_{\text{congestion}}.$$

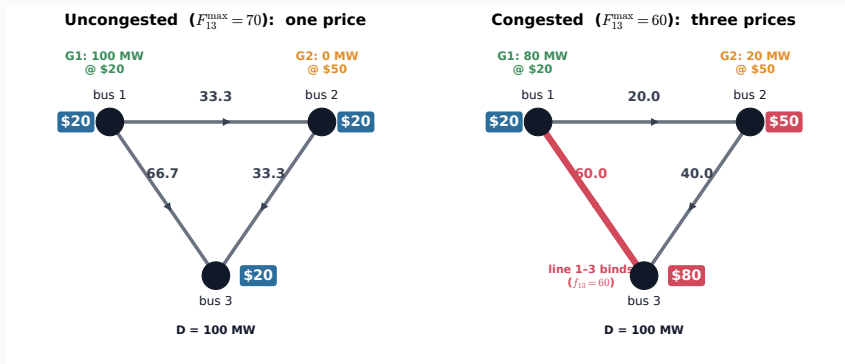
- **Node–arc:** carry n duals λ_i , read price directly. Natural when you want a price *at every bus*.
- **Reduced/PTDF:** carry $1 + |\mathcal{L}_{\text{bind}}|$ duals, *assemble* the price. Natural when only a few lines bind.
- The PTDF column of a binding line = the sensitivity of the node potentials to that line's μ_{ℓ} — the spatial fingerprint of Part II is the *same* object from either side.

Take-away

Pick the bookkeeping that fits the question; the prices are identical. We verify this on the toy next.

Step 4 — the toy PRIMAL, and what the network does

Triangle, equal reactances ($b = 10$). G1@bus1 (\$20), G2@bus2 (\$50), $D_3 = 100$ MW. The current divider gives $\text{flow}_{1 \rightarrow 3} = (200 - P_2)/3$.



Left: with a slack limit the cheapest dispatch (100, 0) is feasible, line 1–3 carries $66.7 < 70$, one price. Right: tightening to 60 forces $P_2 \uparrow 20$, line 1–3 binds (red), and the three buses separate into \$20 / \$50 / \$80.

Take-away

One scalar constraint moves dispatch $(100, 0) \rightarrow (80, 20)$ and turns one price into three. Now price it two ways.

Step 5'' — the toy DUAL in network-flow form

Congested case. Both gens interior $\Rightarrow \lambda_1 = C'_1 = \20 , $\lambda_2 = C'_2 = \$50$ *immediately*. Lines $\{12, 13, 23\}$, only 1–3 binds ($\mu_{13}^+ = \mu$). Uncongested lines: $\eta_\ell = \lambda_{\text{fr}} - \lambda_{\text{to}}$.

$$\eta_{12} = \lambda_1 - \lambda_2 = -30, \quad \eta_{23} = \lambda_2 - \lambda_3, \quad \eta_{13} = (\lambda_1 - \lambda_3) + \mu.$$

Dual KVL on the triangle, $\mathbf{AB}_d \boldsymbol{\eta} = \mathbf{0}$, collapses to $\eta_{12} = \eta_{23} = -\eta_{13}$. Close the loop:

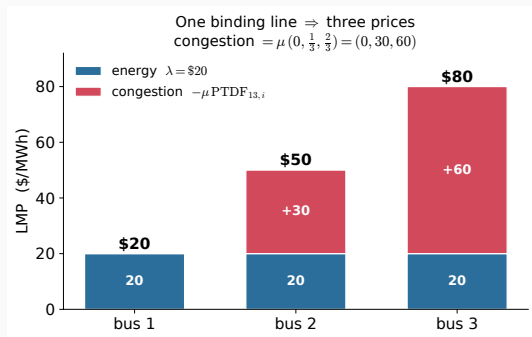
$$\eta_{23} = \eta_{12} \Rightarrow \lambda_2 - \lambda_3 = -30 \Rightarrow \lambda_3 = \$80, \quad \eta_{13} = -\eta_{12} = 30 \Rightarrow (20 - 80) + \mu = 30 \Rightarrow \mu = 90.$$

Node potentials $(\lambda_1, \lambda_2, \lambda_3) = (20, 50, 80)$ = the PTDF answer $\lambda - \mu H_{1-3,i}$ of Step 5 — **identical**.

Take-away

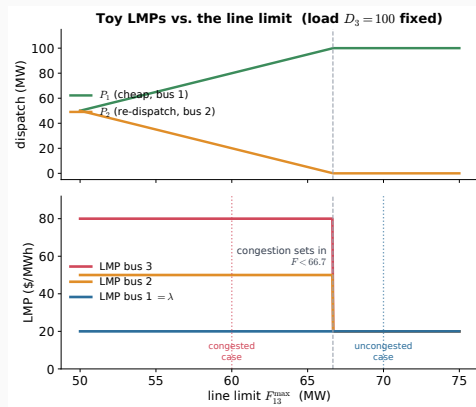
The price at bus 3 falls out of **dual KVL** — no PTDF row needed. Same \$80, derived from the loop's potential balance.

Reading the toy result



Energy \$20 is the floor at every bus; the congestion stack

$= \mu(0, \frac{1}{3}, \frac{2}{3}) = (0, 30, 60)$ is μ times the PTDF column of the binding line.



Sweep $F_{13}^{\max}$ with $D_3 = 100$ fixed. Dispatch moves *continuously*; the LMPs are **piecewise constant** — flat at \$20 while uncongested, then a discrete fan-out the instant line 1 — 3 binds at 66.7 MW.

Take-away

Congestion price is a **step**, not a slope: it switches on with the binding set, then is pinned by *which* units are marginal. This discreteness is precisely term (c) that Part IV will detect.

Triangle, equal reactances $x = 0.1$ pu ($b = 10$ each). Gen 1 @ bus 1 ($c_1 = \$20$), gen 2 @ bus 2 ($c_2 = \50), load $D_3 = 100$ MW @ bus 3. The only flow we need is line 1–3; the current divider of Session 1 gives $\text{flow}_{1 \rightarrow 3} = (200 - P_2)/3$. The LP is:

$$\min_{P_1, P_2} 20P_1 + 50P_2 \quad \text{s.t.} \quad \begin{cases} P_1 + P_2 = 100 & (\lambda) \\ \frac{200 - P_2}{3} \leq F_{13}^{\max} & (\mu) \\ 0 \leq P_1, P_2 \leq 200. \end{cases}$$

- **Uncongested** ($F_{13}^{\max} = 70$): cheapest is all from bus 1, $\hat{P}_1 = 100$, $\hat{P}_2 = 0$; flow = $66.7 < 70$, so $\mu = 0$.
- **Congested** ($F_{13}^{\max} = 60$): the constraint forces $\frac{200 - P_2}{3} \leq 60 \Rightarrow P_2 \geq 20$, so $\hat{P}_1 = 80$, $\hat{P}_2 = 20$ and line 1–3 binds.

Take-away

One scalar constraint changes the optimal dispatch from (100, 0) to (80, 20). Now price it.

The toy DUAL

Congested case. Both generators interior, so only λ and μ are active. Lagrangian:

$$\mathcal{L} = 20P_1 + 50P_2 - \lambda(P_1 + P_2 - 100) + \mu\left(\frac{200-P_2}{3} - 60\right).$$

Stationarity (note $\partial \text{flow}_{13}/\partial P_1 = 0$, $\partial \text{flow}_{13}/\partial P_2 = -\frac{1}{3}$):

$$\frac{\partial \mathcal{L}}{\partial P_1} = 20 - \lambda = 0 \Rightarrow \lambda = 20, \quad \frac{\partial \mathcal{L}}{\partial P_2} = 50 - \lambda - \frac{\mu}{3} = 0 \Rightarrow \mu = 90.$$

PTDF row of line 1–3 (slack = bus 1): $\mathbf{H}_{1-3,:} = (0, -\frac{1}{3}, -\frac{2}{3})$. Apply the boxed formula $\text{LMP}_i = \lambda - \mu H_{1-3,i}$:

bus	λ	$-\mu H_{1-3,i}$	LMP_i	reading
1	20	$-90(0) = \mathbf{0}$	\$20	cheap source, no congestion premium
2	20	$-90(-\frac{1}{3}) = \mathbf{+30}$	\$50	re-dispatched to relieve the line
3	20	$-90(-\frac{2}{3}) = \mathbf{+60}$	\$80	downstream of the constraint — pays most

Take-away

One binding line, three prices. Energy \$20 everywhere; congestion $(0, 30, 60) = \mu$ times the PTDF column. The price map IS a PTDF column.

pandapower DC-OPF, realistic ratings (50 MVA backbone, 16 MVA ring), slack at bus 0. Notebook §1.

bus	LMP _i (€/MWh)	λ_{energy}	congestion
0 (slack)	26.83	26.83	0.000
1	44.60	26.83	+17.77
2	37.08	26.83	+10.24
3	39.43	26.83	+12.59
4	42.63	26.83	+15.80

Binding lines: line 0 (bus 0→1) and line 20 (bus 9→20); largest $|H_{\ell^*,k}|$ at bus 1 (0.833) and bus 20 (0.575).

Take-away

Toy and benchmark are the same calculation at different scale: λ uniform, congestion = $\mu \times$ PTDF.
The bus closest to a binding line pays the most.

The loss component (sketch — why DC-lossless is a choice)

Add a quadratic loss term $P_{\text{loss}}(\boldsymbol{\theta}) \approx \frac{1}{2} \boldsymbol{\theta}^\top \mathbf{G} \boldsymbol{\theta}$ ($\mathbf{G}$ the conductance Laplacian). The KKT then give (Litvinov 2010):

$$\text{LMP}_i = \lambda(1 + \delta_i) - \sum_{\ell} \mu_{\ell} H_{\ell,i}, \quad \delta_i \equiv \frac{\partial P_{\text{loss}}}{\partial P_i}.$$

- δ_i typically $\pm 1\text{--}5\%$ on a stiff transmission grid.
- Small *relative to* congestion here, so we drop it in the lab for clarity; ISOs report the full triple (energy, congestion, loss).
- On distribution feeders this term is *not* negligible (high R/X) — see the aside two slides on.

Stack the per-bus decomposition:

$$\mathbf{p}^{\text{LMP}} = \underbrace{\lambda \mathbf{1}}_{\text{(a) energy}} + \underbrace{\lambda \delta}_{\text{(b) loss}} - \underbrace{\mathbf{H}^{\top} \mu}_{\text{(c) congestion}} .$$

- (a) **spatially constant** — contributes nothing to spatial autocorrelation.
- (b) **smooth** in network position — mild positive dependence.
- (c) **discrete**, can *flip sign* across a binding line — the source of “patchy” LMP maps.

Key idea. Term (c) is why LMP maps are patchy in stressed hours and flat in light hours. Part IV (Moran/LISA) detects the patchiness; Part V (SAR) quantifies it.

Aside — “but what about distribution networks?”

A question you will be asked. Today’s story is *transmission*; distribution differs in three ways:

	Transmission (today)	Distribution (DLMP)
Topology	meshed (loops)	radial (a tree)
R/X ratio	small $\Rightarrow$ DC & lossless OK	large $\Rightarrow$ losses matter
Binding constraint	thermal line limits	often voltage limits first

- The concept survives: the **DLMP** is still a dual, but with *four* components — energy, congestion, **loss**, **voltage**.
- On a radial feeder “neighbour” is just tree adjacency, and $Z_{\text{eff}}(i, j)$ is the sum of impedances along the *unique* path (the toy tree of Session 1).
- Same **W**-based tools apply — but you may *not* drop the loss/voltage terms.

Take-away

Same dual logic, more terms. A clean follow-up project, not a different theory.

III. The spatial weight matrix W

We now stop treating the grid as one circuit and start treating each bus as an **observation** carrying a number y_i (a price, a demand, a curtailment).

Do electrically neighbouring buses tend to carry *similar* values?

Three *mechanistic* reasons grid data are spatially dependent — each one you have already met:

1. **Network coupling.** $P^{\text{inj}} = \mathbf{B}\theta$: an injection anywhere moves angles everywhere.
2. **Shared constraints.** One binding line sets the price at many buses at once, through its PTDF column (Part II).
3. **Common drivers.** Weather / fuel hits a whole region together (often unobserved).

Take-away

To ask the question we need a declared notion of “neighbour”: the **spatial weight matrix** $\mathbf{W}$.

Definition

A **spatial weight matrix** is $\mathbf{W} \in \mathbb{R}^{n \times n}$ with $\mathbf{W}_{ii} = 0$, where $\mathbf{W}_{ij}$ encodes the *a priori* strength of dependence between buses i and j (Anselin and Rey 2014; J. LeSage and Pace 2009).

- Almost always **row-standardised**: $\tilde{\mathbf{W}}_{ij} = \mathbf{W}_{ij} / \sum_k \mathbf{W}_{ik}$, so each row sums to 1 and $(\tilde{\mathbf{W}}\mathbf{y})_i$ is literally the *average of bus i 's neighbours*.
- $\mathbf{W}$ is treated as *known and exogenous* — it is the researcher's declared belief about who interacts with whom.
- Different $\mathbf{W} \Rightarrow$ different answers. *Always run robustness across at least two $\mathbf{W}$.*

Take-away

$\mathbf{W}$ is a modelling choice you must *defend*, not a setting you accept by default.

Path 1–2–3–4. Binary adjacency, then row-standardised:

$$\mathbf{W} = \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{\text{row-std}} \tilde{\mathbf{W}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

- End buses (1, 4) have a single neighbour $\Rightarrow$ weight 1.
- Interior buses split weight $\frac{1}{2}/\frac{1}{2}$.
- $(\tilde{\mathbf{W}}\mathbf{y})_2 = \frac{1}{2}(y_1 + y_3)$: the spatial lag at bus 2 *is* the mean of its two neighbours.

Take-away

Row-standardisation turns “the matrix $\mathbf{W}$ ” into the intuitive operator “replace each value by its neighbours’ average”. Hold this picture for Moran’s I next.

1. **Binary adjacency** $\mathbf{W} = \mathbf{A}$ (row-std). *The naive topological baseline — always report it.*
2. **Electrical distance** $\mathbf{W}_{ij} = 1/Z_{\text{eff}}(i, j)$, with Z_{eff} the effective resistance of Session 1. *Physical; dense; counts all parallel paths.*
3. **PTDF-thresholded** $\mathbf{W}_{ij} = \max_{\ell} |H_{\ell, i} - H_{\ell, j}|$ after quantile thresholding. *The principled choice for nodal prices (Section II told us why).*
4. **Geographic k -NN** when coordinates exist. *Useful when geography drives the data (weather, renewables).*

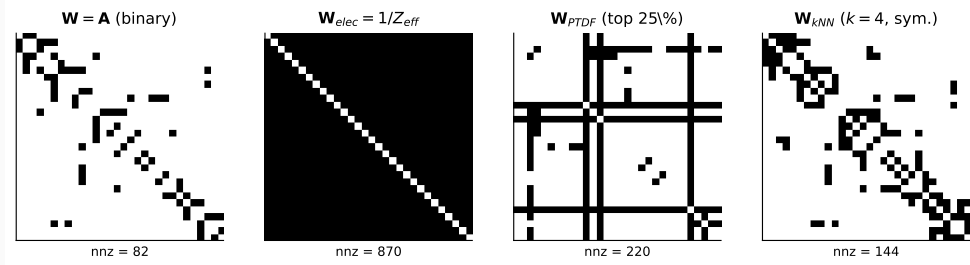
Key idea. Options 2 and 3 are built from objects you already have (Z_{eff} , $\mathbf{H}$). The spatial layer is *not* a new physics — it is the Session-1 circuit objects, re-read as “who is a neighbour”.

Worked example (benchmark) — the four W on IEEE 30

Same 30 buses, four beliefs about neighbourhood. Notebook §1b.1.

Construction	non-zeros	density	max entry
$\mathbf{W} = \mathbf{A}$ (binary)	82	0.091	1.000
$\mathbf{W}_{\text{elec}} = 1/Z_{\text{eff}}$	870	0.967	47.47
$\mathbf{W}_{\text{PTDF}}$ (top 25%)	218	0.242	1.000
$\mathbf{W}_{k\text{NN}}$ ($k = 4$, sym.)	144	0.160	1.000

Four credible neighbourhoods on IEEE 30 (same 30 buses, different beliefs)



Adjacency is sparsest; electrical distance couples every pair; PTDF and k -NN sit between. After row-standardisation all four have spectral radius 1, so the same admissible range $|\rho| < 1$ applies in Part V — but the *meaning* of ρ changes with $\mathbf{W}$.

IV. Detecting dependence — Moran and LISA

We now treat each bus as an **observation** carrying a number y_i (a price, a demand). Two questions, two tools:

Question	Tool
Is there <i>any</i> spatial structure in $\mathbf{y}$?	global Moran's I
<i>Where</i> are the clusters / outliers?	local Moran (LISA)

Take-away

Both need a declared neighbour matrix $\mathbf{W}$ (Part III). Neither controls for covariates — that is what the regression of Part V is for. Detection first, explanation after.

Moran (1950). Centre the data, $\mathbf{z} = \mathbf{y} - \bar{y} \mathbf{1}$:

$$I = \frac{n}{S_0} \frac{\mathbf{z}^\top \mathbf{W} \mathbf{z}}{\mathbf{z}^\top \mathbf{z}}, \quad S_0 = \sum_{ij} \mathbf{W}_{ij}.$$

Algebraic reading. For row-standardised $\mathbf{W}$, $S_0 = n$, so the prefactor is 1 and

$$I = \frac{\mathbf{z}^\top \mathbf{W} \mathbf{z}}{\mathbf{z}^\top \mathbf{z}} = \frac{\sum_i z_i (\mathbf{W} \mathbf{z})_i}{\sum_i z_i^2}.$$

- It is a **Rayleigh quotient** of $\mathbf{W}$: bounded by the eigenvalues of $\mathbf{W}$, $\mu_{\min} \leq I \leq \mu_{\max}$.
- $(\mathbf{W} \mathbf{z})_i$ is bus i 's *neighbour average*: I is the correlation between a value and its neighbours' values.
- Under the no-structure null, $\mathbb{E}[I] = -1/(n-1) \rightarrow 0$.

Take-away

I measures how much $\mathbf{z}$ aligns with the high-eigenvalue directions of $\mathbf{W}$ — the spectral lens of Session 1, reused.

Worked example (toy) — I for the monotone signal, in full

Path 1–2–3–4, row-standardised $\tilde{\mathbf{W}}$. Signal $\mathbf{y} = (1, 2, 3, 4)$, so $\bar{y} = 2.5$ and $\mathbf{z} = (-1.5, -0.5, 0.5, 1.5)$.

1. $\mathbf{z}^\top \mathbf{z} = 1.5^2 + 0.5^2 + 0.5^2 + 1.5^2 = 5.0$.
2. Neighbour averages $\mathbf{Wz}$: $(\mathbf{Wz})_1 = z_2 = -0.5$; $(\mathbf{Wz})_2 = \frac{1}{2}(z_1 + z_3) = -0.5$; $(\mathbf{Wz})_3 = \frac{1}{2}(z_2 + z_4) = +0.5$; $(\mathbf{Wz})_4 = z_3 = +0.5$.
3. $\mathbf{z}^\top \mathbf{Wz} = (-1.5)(-0.5) + (-0.5)(-0.5) + (0.5)(0.5) + (1.5)(0.5) = 2.0$.
4. $I = \frac{2.0}{5.0} = +0.40$.

Take-away

Three lines of arithmetic, no software. Now vary only the *arrangement* of the same numbers.

Worked example (toy) — same numbers, four arrangements

Same $\tilde{\mathbf{W}}$, every signal has mean 2.5:

signal $\mathbf{y}$	$\mathbf{z}^\top \mathbf{W} \mathbf{z}$	I	verdict
(1, 2, 3, 4) monotone	+2.0	+0.40	positive autocorrelation
(1, 1, 4, 4) two clusters	+4.5	+0.50	one clean low high cut
(1, 4, 2, 3) scrambled	−4.5	−0.90	adjacent values jump
(1, 4, 1, 4) alternating	−9.0	−1.00	perfect checkerboard

($\mathbf{z}^\top \mathbf{z} = 5$ for the first/third, 9 for the second/fourth.)

Misconception killed here

“A high or low value is autocorrelated.” **No.** Autocorrelation is a property of $(\mathbf{W}, \mathbf{y})$ *jointly*: identical numbers score +0.50 or −1.00 purely by *arrangement* relative to $\mathbf{W}$.

Plot the spatial lag $(\mathbf{Wz})_i$ (vertical) against z_i (horizontal). The OLS slope of that cloud is

$$\text{slope} = \frac{\sum_i z_i (\mathbf{Wz})_i}{\sum_i z_i^2} = \frac{\mathbf{z}^\top \mathbf{Wz}}{\mathbf{z}^\top \mathbf{z}} = I.$$

- Toy monotone signal: the four points $(z_i, (\mathbf{Wz})_i) = (-1.5, -0.5), (-0.5, -0.5), (0.5, 0.5), (1.5, 0.5)$ lie on a line of slope 0.40.
- The plot splits into four quadrants around the origin — the basis for LISA on the next slide.

Take-away

“Moran’s I ” and “slope of value-vs-neighbour-average” are the same number. The scatterplot is the picture of the formula.

Anselin (1995) split the global I into a per-bus contribution:

$$I_i = \frac{z_i}{m_2} (\mathbf{W}\mathbf{z})_i, \quad m_2 = \frac{1}{n} \sum_k z_k^2.$$

Algebraic link to the global I :

$$\sum_i I_i = \frac{1}{m_2} \sum_i z_i (\mathbf{W}\mathbf{z})_i = \frac{n \mathbf{z}^\top \mathbf{W} \mathbf{z}}{\mathbf{z}^\top \mathbf{z}} = nI \Rightarrow I = \frac{1}{n} \sum_i I_i.$$

Sign of $(z_i, (\mathbf{W}\mathbf{z})_i) \Rightarrow$ quadrant: **HH**/**LL** are clusters, **HL**/**LH** are spatial outliers.

Take-away

The global I is literally the *average* of the local I_i . LISA tells you which buses build it.

Worked example (toy) — LISA for the monotone signal

Same path, $\mathbf{z} = (-1.5, -0.5, 0.5, 1.5)$, $\mathbf{W}\mathbf{z} = (-0.5, -0.5, 0.5, 0.5)$, $m_2 = \frac{5}{4} = 1.25$. Then $l_i = z_i(\mathbf{W}\mathbf{z})_i / m_2$:

bus	z_i	$(\mathbf{W}\mathbf{z})_i$	l_i	quadrant
1	-1.5	-0.5	+0.6	LL (low, low neighbours)
2	-0.5	-0.5	+0.2	LL
3	+0.5	+0.5	+0.2	HH
4	+1.5	+0.5	+0.6	HH (high, high neighbours)

Check: $\sum_i l_i = 1.6$, and $I = 1.6/4 = 0.40$ — the global value, recovered. No outliers (no HL/LH): the monotone signal is two clean clusters.

Take-away

The endpoints (buses 1, 4) contribute most: extreme values sitting among like neighbours drive global autocorrelation.

Worked example (benchmark) — the same detector on prices

Apply Moran to the 200-h IEEE 30 LMP panel, hour by hour (row-standardised $\mathbf{W}$). Notebook §2a.2.

$\mathbf{W}$ choice	mean I across hours	std dev
$\mathbf{W} = \mathbf{A}$	+0.066	0.048
$\mathbf{W} = \mathbf{W}_{\text{PTDF}}$	−0.005	0.005

Headline: $\text{corr}(|I_t|, \# \text{binding lines}_t) = +0.867$.

Key idea. Uncongested hours $\Rightarrow$ the LMP map is $\lambda \mathbf{1}$ and $I \rightarrow 0$; stressed hours $\Rightarrow$ high I . The same four-line calculation you did on the toy, run per hour, becomes a **congestion thermometer**.

V. Explaining dependence — spatial regression, developed

Detection said “yes” — four questions it cannot answer

Moran/LISA detect and localise. The questions a planner actually has need **regression**:

Power-system question	Tool
What drives the LMP, controlling for covariates $\mathbf{X}$?	β in SAR/SEM/SDM
If I add +50 MW capacity at bus j , what happens at <i>other</i> buses?	indirect effect
Where should I site new generation?	effect-ranked index (Part VI)
Is clustering caused by the <i>network</i> or by <i>common weather</i> ?	SAR vs. SEM (LM test)

Take-away

Each question maps to a specific quantity we will now build: a coefficient, a spillover, or a model choice.

The three models — full names, equations, mechanisms

Running example: $\mathbf{y}$ = LMP (€/MWh), $\mathbf{X}$ = (demand, capacity) in MW.

Acronym	Full name	Equation
SAR	<i>Spatial Auto-Regressive (lag)</i>	$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \varepsilon$
SEM	<i>Spatial Error Model</i>	$\mathbf{y} = \mathbf{X} \beta + \mathbf{u}, \mathbf{u} = \lambda \mathbf{W} \mathbf{u} + \varepsilon$
SDM	<i>Spatial Durbin Model</i>	$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \mathbf{W} \mathbf{X} \theta + \varepsilon$

Mechanism each one claims, in the LMP language of Part II:

- **SAR:** *my LMP reacts to my neighbours' LMP* — arbitrage/congestion rent equalising adjacent prices.
- **SEM:** *my unobserved shock correlates with my neighbour's* — a regional weather/fuel driver hidden in $\mathbf{u}$.
- **SDM:** *my neighbours' covariates move my LMP* — a new generator next door, entering through its PTDF (the dual story made statistical).

Take-away

Choosing a model is a *claim about the mechanism*, not an R^2 contest.

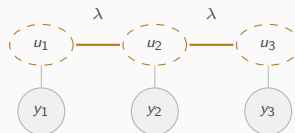
The three models — where does the neighbour enter?

SAR $y = \rho W y + X \beta + u$



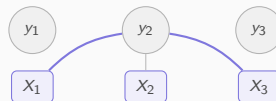
el vecino entra por y

SEM



el vecino entra por el error u

SDM



el vecino entra por X

Take-away

Choosing a model is a *claim about the mechanism*, not an R^2 contest: SAR couples the observed price, SEM the hidden shock, SDM the neighbour's covariate.

Solve SAR for $\mathbf{y}$:

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1}(\mathbf{X}\beta + \varepsilon) \equiv \mathbf{S}_\rho(\mathbf{X}\beta + \varepsilon), \quad \mathbf{S}_\rho = \sum_{k \geq 0} (\rho \mathbf{W})^k.$$

Toy path, $\tilde{\mathbf{W}}$, $\rho = 0.5$:

$$\mathbf{S}_\rho = \begin{bmatrix} 1.156 & 0.622 & 0.178 & 0.044 \\ 0.311 & 1.244 & 0.356 & 0.089 \\ 0.089 & 0.356 & 1.244 & 0.311 \\ 0.044 & 0.178 & 0.622 & 1.156 \end{bmatrix}, \quad \mathbf{S}_\rho \mathbf{1} = \frac{1}{1-0.5} \mathbf{1} = 2 \mathbf{1}.$$

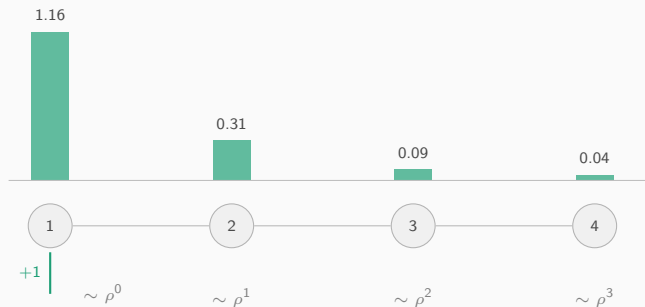
- The Neumann series: term k is the k -hop spillover. Entry $(1, 4) = 0.044 \neq 0$ — a shock 3 hops away still arrives.
- This is the statistical twin of the network reach $\mathbf{B}^{-1}$ from Part II: “a shock anywhere reaches everywhere”.

Take-away

$\mathbf{S}_\rho$ is the regression's version of the DC power-flow multiplier. Hold it — it returns in the bias, the estimator, and the effects.

SAR reduced form — a shock reaches everywhere, with decay

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1}(\mathbf{X}\beta + \varepsilon) \equiv \mathbf{S}_\rho(\mathbf{X}\beta + \varepsilon), \quad \mathbf{S}_\rho = \sum_{k \geq 0} (\rho \mathbf{W})^k.$$



Take-away

Column 1 of $\mathbf{S}_\rho$: a +1 injection at bus 1 still reaches bus 4 ($0.044 \neq 0$). $\mathbf{S}_\rho$ is the statistical twin of the power-flow multiplier $\mathbf{B}^{-1}$ — a shock anywhere reaches everywhere.

Why OLS fails — the bias, derived on the toy

Take pure SAR $\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \varepsilon$, so $\mathbf{y} = \mathbf{S}_\rho \varepsilon$ and $\mathbf{W}\mathbf{y} = \mathbf{W}\mathbf{S}_\rho \varepsilon$. The OLS regressor $\mathbf{W}\mathbf{y}$ is correlated with the error:

$$\mathbb{E}[(\mathbf{W}\mathbf{y})^\top \varepsilon] = \sigma^2 \text{tr}(\mathbf{W}\mathbf{S}_\rho) \neq 0.$$

On the toy path ($\rho = 0.5$): $\text{tr}(\mathbf{W}\mathbf{S}_\rho) = 1.6 > 0$.

- Positive $\Rightarrow \mathbf{W}\mathbf{y}$ and ε move together $\Rightarrow$ OLS on $\mathbf{W}\mathbf{y}$ is biased **upward**.
- The bias is *inconsistent*: it does not vanish as $n \rightarrow \infty$. (Session 1 Monte Carlo: $\rho_0 = 0.5$ on IEEE 30 gave $\hat{\rho}_{\text{OLS}} = 0.70$, bias $+0.20$.)

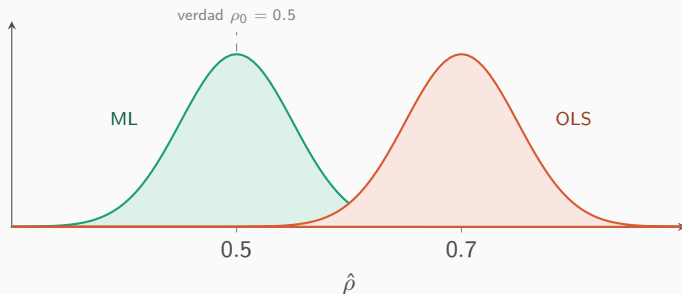
Take-away

The endogeneity is not a nuisance — it is the same “reaches everywhere” property of $\mathbf{S}_\rho$, now poisoning OLS. We need an estimator that knows about $\mathbf{S}_\rho$.

Why OLS fails — biased *and* inconsistent

Pure SAR $\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \varepsilon \Rightarrow \mathbf{W}\mathbf{y}$ is correlated with ε :

$$\mathbb{E}[(\mathbf{W}\mathbf{y})^\top \varepsilon] = \sigma^2 \text{tr}(\mathbf{W}\mathbf{S}_\rho) = 1.6 > 0 \Rightarrow \hat{\rho}_{\text{OLS}} \text{ biased upward.}$$



Take-away

The bias is *inconsistent*: numerator and denominator both scale with n , so $\hat{\rho}_{\text{OLS}} \rightarrow 0.7$ no matter how many buses you add. More data cannot save OLS — ML's Jacobian must.

Maximum likelihood maximises the concentrated log-likelihood

$$\ell(\rho) = \text{const} + \underbrace{\log|\mathbf{I} - \rho\mathbf{W}|}_{\text{Jacobian}} - \frac{n}{2} \log \hat{\sigma}^2(\rho),$$

and the Jacobian is a sum over the eigenvalues μ_k of $\mathbf{W}$:

$$\log|\mathbf{I} - \rho\mathbf{W}| = \sum_k \log(1 - \rho\mu_k).$$

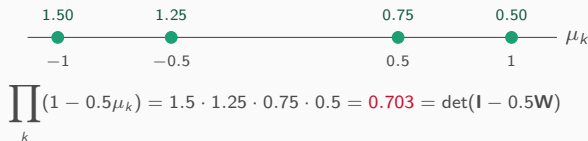
Toy path: $\mu_k \in \{-1, -\frac{1}{2}, +\frac{1}{2}, +1\}$, so at $\rho = 0.5$

$$\log|\mathbf{I} - 0.5\mathbf{W}| = \log(1.5) + \log(1.25) + \log(0.75) + \log(0.5) = \log(0.703).$$

- The Jacobian is exactly the term OLS omits — it “credits” the spatial feedback so $\hat{\rho}$ becomes consistent.
- Same eigenvalues bound ρ ($|\rho| < 1/\mu_{\max} = 1$) and set Moran’s Rayleigh quotient. *One spectrum, three uses.*

$$\ell(\rho) = \text{const} + \underbrace{\log |\mathbf{I} - \rho \mathbf{W}|}_{\text{Jacobian}} - \frac{n}{2} \log \hat{\sigma}^2, \quad \log |\mathbf{I} - \rho \mathbf{W}| = \sum_k \log(1 - \rho \mu_k).$$

factor $(1 - \rho \mu_k)$ at $\rho = 0.5$

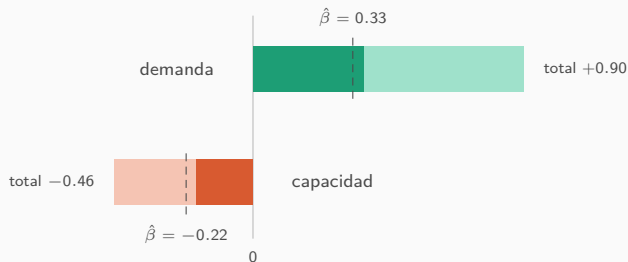


Take-away

One spectrum, three uses: it bounds ρ ($|\rho| < 1/\mu_{\max} = 1$), sets Moran's Rayleigh quotient, and builds this Jacobian — the term OLS omits.

Direct / indirect / total — $\hat{\beta}$ is *not* the marginal effect

$$\mathbf{S}_r = \beta_r \mathbf{S}_\rho : \quad \text{Direct} = \frac{1}{n} \text{tr} \mathbf{S}_r, \quad \text{Total} = \frac{1}{n} \mathbf{1}^\top \mathbf{S}_r \mathbf{1} = \frac{\beta_r}{1-\rho}, \quad \text{Indirect} = \text{Total} - \text{Direct}.$$



■ / ■ directo ■ / ■ indirecto (spillover) teal = sube precio, coral = baja. OLS reporta 0.33; el total real es 0.90 ($\sim 2.7\times$).

Take-away

$\hat{\beta}$ is neither the own-bus effect (0.37) nor the system total (0.90): the spillover (indirect) is invisible to OLS.

Since $\mathbf{y} = \mathbf{S}_\rho(\mathbf{X}\beta + \varepsilon)$, the effect of regressor r is the *matrix* $\mathbf{S}_r = \beta_r \mathbf{S}_\rho$. Summaries (J. LeSage and Pace 2009):

$$\text{Direct} = \frac{1}{n} \text{tr}(\mathbf{S}_r), \quad \text{Total} = \frac{1}{n} \mathbf{1}^\top \mathbf{S}_r \mathbf{1} = \frac{\beta_r}{1-\rho}, \quad \text{Indirect} = \text{Total} - \text{Direct}.$$

Toy path, $\rho = 0.5$, $\beta_{\text{dem}} = 0.30$ €/MWh per MW. $\text{diag}(\mathbf{S}_\rho) = (1.156, 1.244, 1.244, 1.156)$:

$$\text{Direct} = 0.30 \cdot \frac{1}{4} (1.156 + 1.244 + 1.244 + 1.156) = 0.36,$$

$$\text{Total} = \frac{0.30}{1-0.5} = 0.60, \quad \text{Indirect} = 0.60 - 0.36 = 0.24.$$

Take-away

$\hat{\beta} = 0.30$ is **neither** the own-bus effect (0.36) **nor** the system total (0.60): 40% of the marginal effect lands *off* the host bus. Reporting $\hat{\beta}$ alone is wrong.

Worked example (synthetic) — a SAR fit you can interpret

IEEE 30 LMPs are too flat to teach interpretation, so we use a **synthetic 64-bus mesh** (8×8 row-standardised lattice). **Known DGP:** $\mathbf{y} = \rho_0 \mathbf{W}\mathbf{y} + \beta_0 + \beta_{\text{dem}}\text{dem} + \beta_{\text{cap}}\text{cap} + \varepsilon$, with $\rho_0 = 0.60$, $\beta_{\text{dem}} = +0.30$, $\beta_{\text{cap}} = -0.15$, $\sigma = 2$, seed 2026.

Parameter	OLS (<i>wrong model</i>)	ML SAR (<i>correct</i>)	truth
$\hat{\rho}$	—	0.650	0.60
$\hat{\beta}_{\text{dem}}$	0.325	0.313	0.30
$\hat{\beta}_{\text{cap}}$	-0.221	-0.160	-0.15
intercept	36.85	2.32	5.0

- **Read $\hat{\rho} = 0.65$:** if my neighbours' average LMP rises €1/MWh, mine rises €0.65/MWh *at once*, controlling for $\mathbf{X}$.
- **Read $\hat{\beta}_{\text{dem}} = 0.31$:** the partial own-bus slope, *before* the network propagates it.
- **OLS damage:** it has no ρ , dumps the feedback into the intercept (36.9 vs. 5), and would call $\hat{\beta}_{\text{dem}} \approx 0.33$ “the marginal effect”.

Worked example (synthetic) — SAR effects, interpreted

From $\mathbf{S}_\rho = (\mathbf{I} - 0.65\mathbf{W})^{-1}$ on the same fit:

regressor	direct	indirect	total
demand (€/MWh per MW)	+0.367	+0.528	+0.895
capacity (€/MWh per MW)	−0.188	−0.270	−0.458

- **Demand:** +1 MW of load raises the host LMP by €0.37/MWh and adds €0.53/MWh across the rest of the mesh — total €0.90/MWh.
- OLS reported a “marginal effect” of ≈ 0.33 . The **true total is 0.90 — OLS understates it $\sim 2.7\times$** by ignoring $\mathbf{S}_\rho$.
- **Capacity:** +1 MW lowers LMPs by €0.46/MWh in total — again the spillover (−0.27) exceeds the own effect (−0.19).

Take-away

This is the answer to “what happens at *other* buses?” — the indirect column. It is invisible to OLS.

The effect matrix $S = \beta S_\rho$ — one figure ties V to V_I

		emite j				fila =
recibe i		1.16	0.62	0.18	0.04	2.0
		0.31	1.24	0.36	0.09	2.0
		0.09	0.36	1.24	0.31	2.0
		0.04	0.18	0.62	1.16	2.0
col =		1.6	2.4	2.4	1.6	

- **Diagonal** (boxed) = **direct** effect.
- **Row sum** = 2.0 always: every bus *receives* the same (row-standardisation).
- **Column sum** (1.6, 2.4, 2.4, 1.6) *varies*: every bus *emits* differently.

Key idea. Siting picks an emitter j : what matters is the **column sum** (Part VI), not the diagonal. One matrix, two readings.

Worked example (synthetic) — SDM: own vs. neighbour covariates

Same mesh, now the DGP also has **neighbour-covariate** terms $\mathbf{WX}\theta$ with $\theta_{\text{dem}} = +0.10$, $\theta_{\text{cap}} = -0.20$.
Fitting the Spatial Durbin Model:

regressor	own $\hat{\beta}$	nbr $\hat{\theta}$	direct	indirect
demand	+0.285	+0.083	+0.351	+0.656
capacity	-0.159	-0.178	-0.229	-0.692

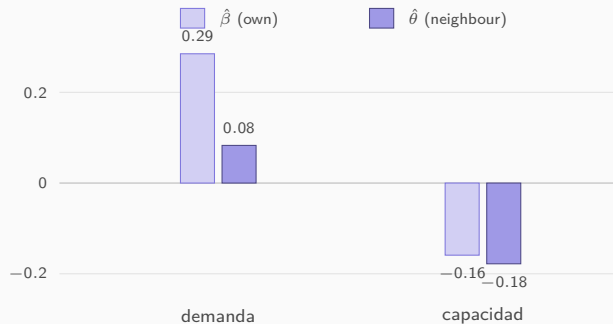
($\hat{\rho} = 0.635$; truth $\rho_0 = 0.60$, $\beta_0 = (0.30, -0.15)$, $\theta_0 = (0.10, -0.20)$ — all recovered.)

- $\hat{\theta}_{\text{cap}} = -0.18$: a generator *next door* pushes *my* LMP down directly, before any feedback — this is the SDM-only term.
- Result: capacity's **indirect effect -0.69** is **3× its direct -0.23**. A +1 MW capacity addition helps the *region* far more than the host bus.

Take-away

SDM separates “effect on me of *my* $\mathbf{X}$ ” from “effect on me of *my neighbours'* $\mathbf{X}$ ” — exactly the decomposition a siting planner needs (Part VI).

SDM — separating my own X from my neighbour's X



Take-away

$\hat{\theta}_{\text{cap}} = -0.18$: a generator *next door* pushes my LMP down directly. Capacity's indirect effect ends up $3\times$ its direct one — exactly the decomposition a siting planner needs (Part VI).

Which model? Diagnose, don't eyeball R^2

After OLS, the Lagrange-multiplier battery (Anselin, Bera, et al. 1996) arbitrates (the *robust* forms net out cross-contamination):

Robust LM-lag	Robust LM-error	Pick
significant	not	SAR (network contagion)
not	significant	SEM (common regional shock)
both	both	SDM (use the Durbin lag)

J. P. LeSage (2008): under doubt, **SDM is the safe default** — it stays consistent for the effects of interest under either omitted spatial covariates or correlated unobservables.

Take-away

Answer to “network or weather?” — the LM-lag vs. LM-error contrast. It is a hypothesis test on the *mechanism*.

VI. DG siting as inverse regression

- Parts II–V **described** the grid: $\mathbf{y}$ was a price, and regression told us what drives it and how shocks spill over.
- Now we **prescribe**. Let $\mathbf{y} = \mathbf{s}$ be a **suitability index** for siting distributed generation (DG) at each bus. The same SAR machinery ranks *where* to invest — with network spillovers (Faria et al. 2016; Pereira and Saraiva 2011).

Key idea. This is **inverse regression**: we do not estimate the effect of a cause we observe; we use the estimated effect matrix $\mathbf{S} = \beta \mathbf{S}_\rho$ to score an intervention we are *choosing* — a DG at bus j .

Take-away

The question: “*Where should new generation go?*” — the third of the four questions opened in Part V.

VI. DG siting as inverse regression

From describing prices to prescribing sites — “inverse” regression

Part V — regression (describe)

observe $\mathbf{y}, \mathbf{X}$  $\hat{\rho}, \hat{\beta}$

given an effect $\Rightarrow$ estimate β

Part VI — inverse regression (prescribe)

$\hat{\beta} +$ choose bus j  score for every bus

given $\beta \Rightarrow$ choose bus j

Key idea. No estimation and no matrix inversion are new: it is the same effect matrix $\mathbf{S} = \hat{\beta}\mathbf{S}_p$ from Part V. Only the **question** changes — from “which β ?” to “which bus j ?”.

Take-away

The third of Part V’s four questions: *where should new generation go?*

Fit a SAR on the index s :

$$s_i = \rho \sum_j W_{ij} s_j + \beta_0 + \beta_b b_i + \beta_D D_i + \beta_E E_i + \beta_r r_i + \varepsilon_i.$$

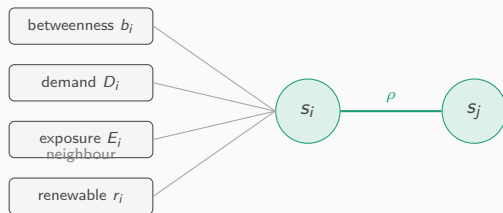
Regressor	Why it belongs
Betweenness b_i	buses on many shortest paths relieve binding lines disproportionately
Local demand D_i	DG near load cuts feeder losses, frees upstream capacity
Congestion exposure E_i	how often bus i sits on a binding line (the Lab 2a output)
Renewable potential r_i	resource proxy from bus coordinates

Take-away

The principled $\mathbf{W}$ here is $\mathbf{W}_{\text{PTDF}}$ (Part III): siting value travels along the same PTDF columns that set the LMP.

The suitability index — regressors and their reach

$$s_i = \rho \sum_j W_{ij} s_j + \beta_0 + \beta_b b_i + \beta_D D_i + \beta_E E_i + \beta_r r_i + \varepsilon_i$$



- b_i : buses on many shortest paths relieve more lines.
- D_i : DG near load cuts losses, frees upstream capacity.
- E_i : how often the bus sits on a binding line.
- r_i : renewable potential (from coordinates).

Key idea. The principled $\mathbf{W}$ is $\mathbf{W}_{\text{PTDF}}$: a site's value travels along the same PTDF columns that set the LMP.

From the SAR reduced form $\mathbf{s} = \mathbf{S}_\rho(\mathbf{X}\beta + \varepsilon)$, the effect of regressor r is the matrix $\mathbf{S}_r = \beta_r \mathbf{S}_\rho$, with

$$(\mathbf{S}_r)_{ij} = \beta_r (\mathbf{S}_\rho)_{ij} = \frac{\partial s_i}{\partial x_{j,r}}.$$

Adding DG at bus j raises $x_{j,r}$ by one unit, producing the **column** $\mathbf{S}_r[:,j]$. Two ways to score that site:

$$\underbrace{(\mathbf{S}_r)_{jj} = \beta_r (\mathbf{S}_\rho)_{jj}}_{\text{own / myopic (ADE-style)}} \quad \text{vs.} \quad \underbrace{\sum_i (\mathbf{S}_r)_{ij} = \beta_r \sum_i (\mathbf{S}_\rho)_{ij}}_{\text{social / total (ATE-style)}}.$$

Key idea. For row-standardised $\mathbf{W}$, the *row* sums of $\mathbf{S}_\rho$ are constant ($1/(1 - \rho)$), but the *column* sums vary across buses. So the social value of a site is a property of its **network position** — and the host-only score misses it.

Worked example (toy) — own value vs. social value

Path 1–2–3–4, $\mathbf{S}_\rho = (\mathbf{I} - 0.5\mathbf{W})^{-1}$ (the same matrix as Part V). Compare its diagonal to its column sums:

bus	1	2	3	4
diagonal $(\mathbf{S}_\rho)_{jj}$ (own value)	1.156	1.244	1.244	1.156
column sum $\sum_i (\mathbf{S}_\rho)_{ij}$ (social)	1.600	2.400	2.400	1.600
row sum (constant)	2.0	2.0	2.0	2.0

- Own value barely separates interior buses (2,3) from ends (1,4): 1.244 vs. 1.156 — a 7% gap.
- Social value separates them sharply: 2.400 vs. 1.600 — a 50% gap. Intervening at an interior bus reaches far more of the network.

Take-away

A myopic planner ranking by own value treats all four buses as near-equal. The social value says the *interior* buses are network-pivotal. Same $\mathbf{S}_\rho$, two very different verdicts.

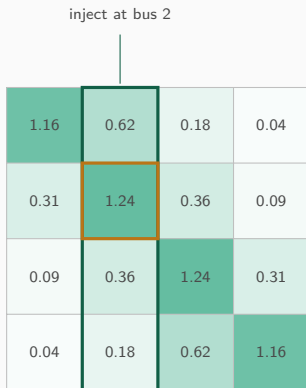
How to value a site — read the *column* of $\mathbf{S}$

$$(\mathbf{S}_r)_{ij} = \beta_r(\mathbf{S}_\rho)_{ij} = \frac{\partial s_i}{\partial x_{j,r}}$$

$\underbrace{(\mathbf{S}_r)_{ij}}$
own (diagonal)

vs.

$\underbrace{\sum_i (\mathbf{S}_r)_{ij}}$
social (column sum)



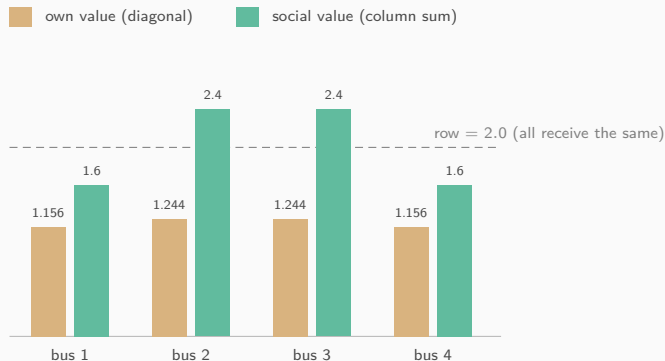
$\Sigma = 2.4$

(own 1.24)

Adding DG at bus j produces the **column** $\mathbf{S}[:,j]$:
how *every* bus reacts.

- **diagonal** = benefit to the host bus itself.
- **column sum** = total benefit across the network.

Receiver vs. sender — the asymmetry behind the disclaimer



Take-away

As *receivers* all buses are identical (row = 2.0). As *emitters* they differ: own value separates interior from ends by 7%; social value by **50%**. Siting picks an emitter $\Rightarrow$ the column rules, not the diagonal.

Row-standardised $\mathbf{W} = \mathbf{A}$, ML. Notebook §2b.2 (pandapower).

Variable	Coefficient	z
constant	−0.923	−6.59
betweenness	+0.377	+1.06
demand density	+0.000	+0.08
congestion exposure	+1.728	+12.09
renewable potential	+0.321	+1.60
ρ (spatial lag)	+0.727	—

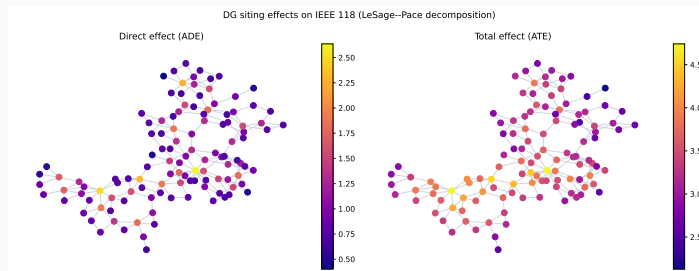
- Read $\hat{\beta}_E = 1.73$: one extra unit of congestion exposure raises a bus's own suitability by 1.73 *before* the network responds.
- Read $\hat{\rho} = 0.73$: strong contagion — a suitable bus makes its neighbours look suitable too.
- Total effect of exposure: $\beta_E / (1 - \rho) = 1.728 / 0.273 \approx 6.33$, i.e. $3.7\times$ the raw $\hat{\beta}_E$.

Worked example (benchmark) — effects and the ranking shift

Per-regressor effects from $\mathbf{S}_\rho = (\mathbf{I} - 0.727\mathbf{W})^{-1}$ (notebook §2b.3):

regressor	direct	indirect	total	% indirect
betweenness	+0.479	+0.901	+1.380	65.3%
congestion exposure	+2.196	+4.130	+6.327	65.3%
renewable potential	+0.409	+0.768	+1.177	65.3%

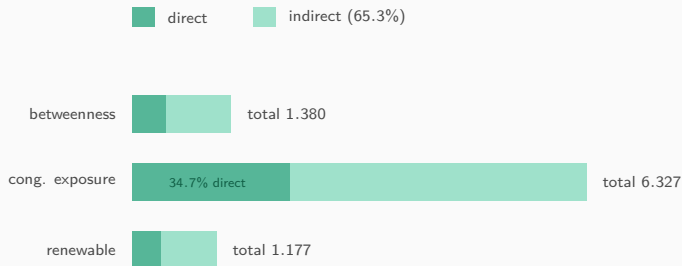
Ranking the 118 buses by host value vs. social value: the top-10 lists share only **6 of 10**. Promoted: {93, 95, 97, 98} (downstream of high-exposure corridors); demoted: {11, 16, 36, 84}.



Left: direct (ADE) per bus. Right: total (ATE) per bus. Notebook §2b.5 (pandapower).

Every regressor splits the same way — the 65.3% explained

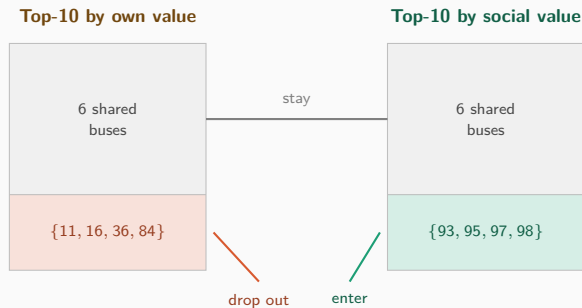
$$\frac{\text{indirect}}{\text{total}} = 1 - (1 - \rho) \frac{1}{n} \text{tr}(\mathbf{S}_\rho) = 1 - 0.347 = \mathbf{0.653} \quad (\text{the } \beta_r \text{ cancels}).$$



Take-away

The direct/indirect split depends only on $(\rho, \mathbf{W})$, not on the regressor: that is why all three share 65.3%. It is a fingerprint of **topology and contagion**.

The ranking flips — own value vs. social value



Take-away

IEEE 118: 6/10 shared. Promoted = interior buses downstream of congested corridors; demoted = high own value but little reach. On the 8×8 lattice the centre-edge contrast is extreme: only 2/10.

VII. Reality check + wrap-up

IEEE 30 and 118 are pedagogically convenient but synthetic. As a sanity check we replay Lab 2a's machinery on an ISO-style benchmark.

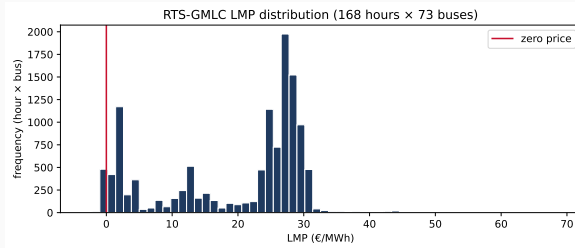
RTS-GMLC (Barrows et al. 2020): NREL's update of the classic RTS-96.

- 73 buses, 105 lines, 15 transformers, 155 generators (Coal, Gas, Nuclear, Hydro, Solar PV, Wind).
- Hourly load for 8760 hours; lat/lng for every bus.
- MIT-licensed; bundled in `notebooks/data/rts_gmlc/`.

168-hour DC-OPF on RTS-GMLC region 1, ratings loosened $1.3\times$ (notebook §3.1). Across $73 \times 168 = 12\,264$ entries:

mean = 19.51, median = 24.94, std = 10.95, range = $[-3.38, +68.51]$ €/MWh.

Negative-price entries: 280 (2.3%) — a documented real-ISO phenomenon in renewable-rich systems (Edmunds et al. 2018).

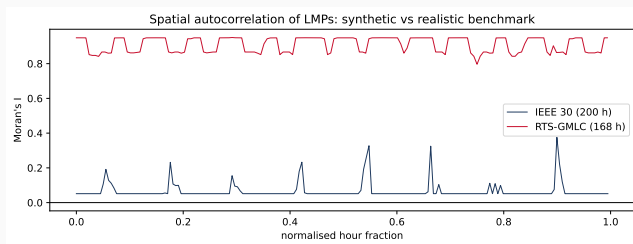


The ρ contrast — synthetic vs. realistic

SAR-MLE on the average LMP:

$$\hat{\rho}_{\text{IEEE 30}} = -0.106 \text{ (NS)}, \quad \hat{\rho}_{\text{RTS-GMLC}} = +0.967 \text{ (highly significant)}.$$

Mean Moran's I : +0.066 vs. +0.910.



Key idea. On a small synthetic grid the spatial structure of LMPs is faint; on a realistic 73-bus benchmark it **dominates**. The pedagogical example was honest; the realistic one is decisive.
Benchmark size and realism matter.

Things you no longer believe

1. "LMP differences are about geography." *No — about which line constraints bind (II).*
2. "A high value is spatially autocorrelated." *No — autocorrelation is a property of $(\mathbf{W}, \mathbf{y})$ jointly (IV).*
3. "Geographic distance is the right $\mathbf{W}$ for nodal prices." *No — the PTDF-derived $\mathbf{W}$ is principled (III, V).*
4. "The raw $\hat{\beta}$ is the marginal effect." *No — use direct/indirect/total (V, VI).*
5. "A small benchmark generalises to a real ISO." *Not always; RTS-GMLC's +0.97 vs. IEEE 30's -0.11 (VII).*

1. LMPs are duals of the DC-OPF balance constraints; spatial structure comes from $-\mathbf{H}^\top \mu$ (Schweppe et al. 1988; Hogan 1992).
2. The 3-bus toy (20/50/80) and IEEE 30 tell the same story: closer to a binding line = higher price.
3. $\mathbf{W}$ is a declared belief built from circuit objects ($Z_{\text{eff}}, \mathbf{H}$); Moran on LMPs is a **congestion thermometer** ($\rho = +0.867$ on IEEE 30).
4. SAR/SEM/SDM map onto network coupling / common shocks / attribute spillover; never read $\hat{\beta}$ as the marginal effect.
5. DG siting is inverse regression; ranking by ATE captures network value OLS misses (Faria et al. 2016). On RTS-GMLC $\hat{\rho}$ swings to +0.97 (Edmunds et al. 2018).

- Centralised DC-OPF assumes truthful bids and complete information. Both fail in real markets (California 2000–2001).
- When they fail, the LMP signal becomes noisy and centralised clearing under-performs decentralised learning.
- Session 3: agents observe local PTDF rows, act on the LMP (the dual we derived today), and the same $\mathbf{W}$ becomes their **communication graph**.
- Same $\mathbf{W}$, same $\mathbf{H}$, same $\mathbf{B}$ — three sessions, one mathematical object.

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Thank you.

Questions → Session 3 (multi-agent simulation and MARL).